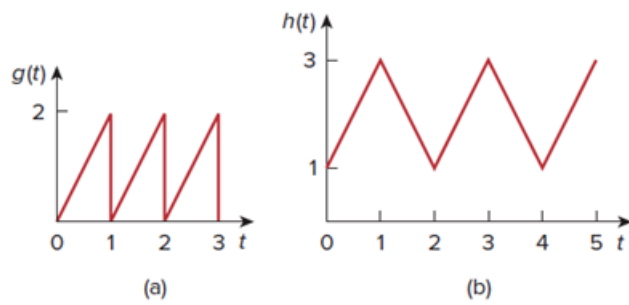


Problem

Find the Laplace transforms of the functions in Fig. 15.34.

Figure 15.34:



Step-by-step solution

Step 1 of 7

(a)

Consider the following signal $g(t)$:

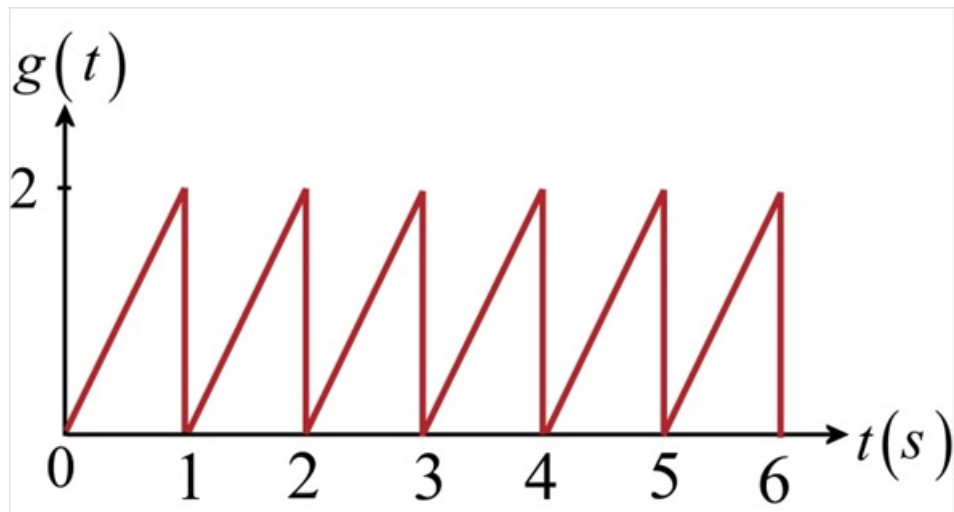


Figure 1

Step 2 of 7

Find the Laplace transform of the signal $g(t)$:

Clearly from Figure 1, $g(t)$ is a periodic signal of time-period 3 sec.

Therefore, Laplace transform of the signal $g(t)$ is $G(s) = \frac{G_1(s)}{1 - e^{-sT}}$.

Substitute 1 for T in this expression.

$$G(s) = \frac{G_1(s)}{1 - e^{-s(1)}} \dots\dots (1)$$

Where, $G_1(s)$ is the lapace transform of $g_1(t)$ which is $g(t)$ over its first period only.

Represent the signal $g_1(t)$ in mathematical form:

$$\begin{aligned} g_1(t) &= 2t[u(t) - u(t-1)] \\ &= 2tu(t) - 2tu(t-1) \\ &= 2tu(t) - 2(t-1+1)u(t-1) \\ &= 2tu(t) - 2(t-1)u(t-1) - 2u(t-1) \end{aligned}$$

Step 3 of 7

Calculate the Laplace transform of the signal $g_1(t)$.

Apply Laplace transform to the signal $g_1(t)$.

$$\begin{aligned} G_1(s) &= \mathcal{L}[2tu(t) - 2(t-1)u(t-1) - 2u(t-1)] \\ &= \mathcal{L}[2tu(t)] - \mathcal{L}[2(t-1)u(t-1)] - \mathcal{L}[2u(t-1)] \\ &= 2\mathcal{L}[tu(t)] - 2\mathcal{L}[(t-1)u(t-1)] - 2\mathcal{L}[u(t-1)] \quad G_1(s) = \frac{2}{s^2}(1 - e^{-s} - se^{-s}) \\ &= 2\left(\frac{1}{s^2}\right) - 2\left(\frac{1}{s^2}e^{-s}\right) - 2\left(\frac{1}{s}e^{-s}\right) \end{aligned}$$

Therefore, the Laplace transform of the signal $g_1(t)$ is $\frac{2}{s^2}(1 - e^{-s} - se^{-s})$.

Step 4 of 7

Substitute $\frac{2}{s^2}(1 - e^{-s} - se^{-s})$ for $G_1(s)$ in equation 1.

$$\begin{aligned} G(s) &= \frac{1}{1 - e^{-s(1)}} \frac{2}{s^2}(1 - e^{-s} - se^{-s}) \\ &= 2 \frac{(1 - e^{-s} - se^{-s})}{s^2(1 - e^{-s})} \end{aligned}$$

Therefore, the Laplace transform of the signal $g(t)$ is $\boxed{2 \frac{(1 - e^{-s} - se^{-s})}{s^2(1 - e^{-s})}}$.

Step 5 of 7

(b)

Consider the following signal $h(t)$:

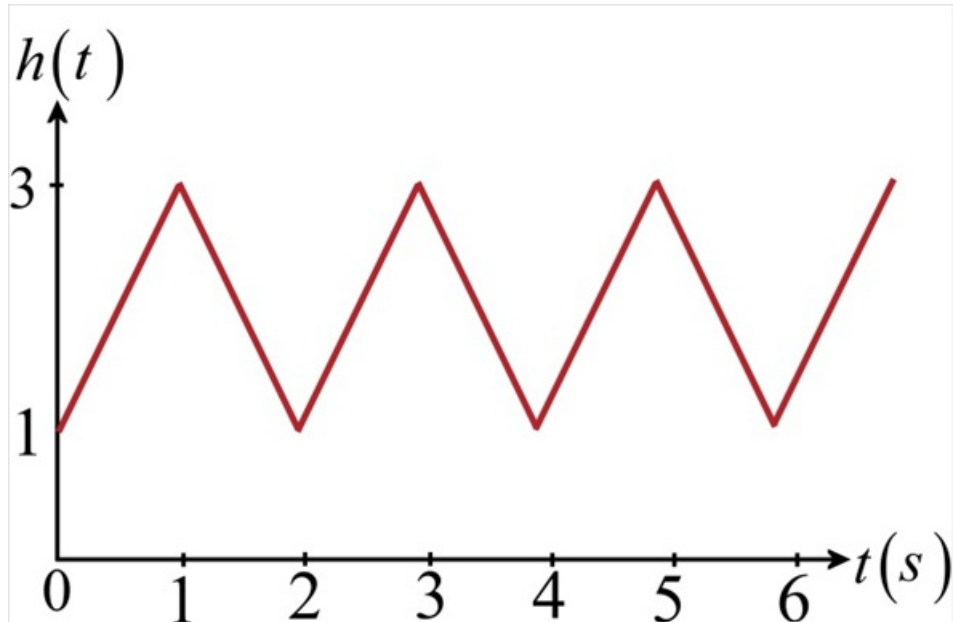


Figure 2

Find the Laplace transform of the signal $h(t)$:

Clearly from Figure 2, $h(t)$ is a periodic signal of time-period 2 sec.

Therefore, Laplace transform of the signal $h(t)$ is $H(s) = \frac{H_1(s)}{1 - e^{-sT}}$.

Substitute 2 for T in this expression.

$$H(s) = \frac{H_1(s)}{1 - e^{-s(2)}} \dots\dots (2)$$

Where, $H_1(s)$ is the lapace transform of $h_1(t)$ which is $h(t)$ over its first period only.

Represent the signal $h_1(t)$ in mathematical form:

$$\begin{aligned} h_1(t) &= (2t+1)[u(t)-u(t-1)] + (-2t+5)[u(t-1)-u(t-2)] \\ &= \left\{ \begin{aligned} &2tu(t) - 2tu(t-1) + [u(t)-u(t-1)] - 2tu(t-1) + \\ &2tu(t-2) + 5[u(t-1)-u(t-2)] \end{aligned} \right\} \\ &= 2tu(t) - 4tu(t-1) + 2tu(t-2) + u(t) + 4u(t-1) - 5u(t-2) \\ &= \left\{ \begin{aligned} &2tu(t) - 4(t-1)u(t-1) - 4u(t-1) + 2(t-2)u(t-2) + \\ &4u(t-2) + u(t) + 4u(t-1) - 5u(t-2) \end{aligned} \right\} \\ h_1(t) &= 2tu(t) - 4(t-1)u(t-1) + 2(t-2)u(t-2) + u(t) - u(t-2) \end{aligned}$$

Calculate the Laplace transform of the signal $h_1(t)$.

Apply Laplace transform to the signal $h_1(t)$.

$$\begin{aligned}
 H_1(s) &= \mathcal{L}[2tu(t) - 4(t-1)u(t-1) + 2(t-2)u(t-2) + u(t) - u(t-2)] \\
 &= \left\{ \mathcal{L}[2tu(t)] - \mathcal{L}[4(t-1)u(t-1)] + \right. \\
 &\quad \left. \mathcal{L}[2(t-2)u(t-2)] + \mathcal{L}u(t) - \mathcal{L}[u(t-2)] \right\} \\
 &= \left\{ 2\mathcal{L}[tu(t)] - 4\mathcal{L}[(t-1)u(t-1)] - \right. \\
 &\quad \left. 2\mathcal{L}[(t-2)u(t-2)] + \mathcal{L}u(t) - \mathcal{L}[u(t-2)] \right\} \\
 &= 2\left(\frac{1}{s^2}\right) - 4\left(\frac{1}{s^2}e^{-s}\right) - 2\left(\frac{1}{s^2}e^{-2s}\right) + \frac{1}{s} - \frac{1}{s}e^{-2s} \\
 H_1(s) &= \frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2}
 \end{aligned}$$

Therefore, the Laplace transform of the signal $h_1(t)$ is $\frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2}$.

Step 7 of 7

Substitute $\frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2}$ for $H_1(s)$ in equation 2.

$$\begin{aligned}
 H(s) &= \frac{1}{1 - e^{-s(2)}} \frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2} \\
 &= \frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2(1 - e^{-2s})}
 \end{aligned}$$

Therefore, the Laplace transform of the signal $h(t)$ is $\boxed{\frac{(2 - 4e^{-s} - 2e^{-2s}) + s(1 - e^{-2s})}{s^2(1 - e^{-2s})}}$.